

Evolving Structures of Interagency Collaboration in Disaster Response: A Social Network Analysis of Cyclone Idai

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Abstract

The complexity of large-scale disasters requires coordinated action among diverse humanitarian organizations. This study examines the interagency collaboration during the response to Cyclone Idai using Social Network Analysis to trace how organizational relationships evolved. Based on ReliefWeb reports, we build temporal networks for early, mid, and late phases of the crisis, linking agencies and reports in a bipartite structure projected into agency-only networks. Metrics such as degree centrality, betweenness, community structure, and role assortativity are used to identify key actors and structural patterns. The results show a shift from centralized coordination led by major UN agencies to more decentralized and specialized collaborations during recovery. Using Complex Adaptive Systems theory, we interpret these changes as signs of emergent, adaptive behavior within the coordination system. The study highlights the importance of inclusive and flexible network structures in disaster governance and emphasizes the joint role of global and local actors in achieving resilient outcomes.

CCS Concepts

• Applied computing; • Computing methodologies; • Social and professional topics;

Keywords

Ubiquitous Computing, Disaster Response, Inter-agency Collaboration, Complex Adaptive Systems, Coordination Networks

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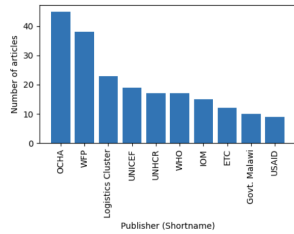
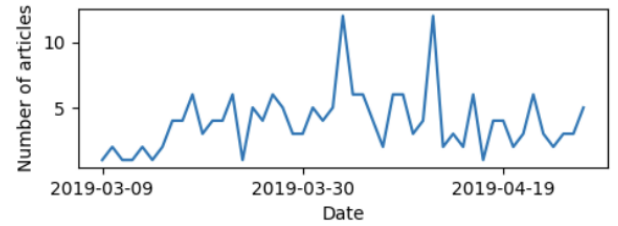
1 Introduction

Effective disaster response relies on the coordinated action of diverse organizations operating under high uncertainty [16]. Government agencies, international bodies, NGOs, and community-based actors must bridge structural and procedural differences to collaborate effectively. Interorganizational networks form the foundation for such collaboration, facilitating information sharing, logistical alignment, and resource allocation. As climate-induced hazards intensify, the need for adaptive and cross-jurisdictional coordination becomes increasingly critical [5, 12]. Empirical evidence underscores the value of collaborative structures in enhancing situational awareness, minimizing redundancies, and improving agility. For instance, the 2013 Boston Marathon bombing illustrated the benefits of horizontal coordination among multiple government levels [13]. In contrast, South Korea's decentralized flood response enabled flexible adaptation [10], while China's centralized model offered command efficiency but limited local responsiveness [14]. These cases highlight how the structure and dynamics of interagency collaboration directly shape disaster response outcomes [11, 18]. However, structural arrangements alone do not guarantee effective collaboration. Elvegård et al. [8] argue that collaboration competence, defined as the ability to engage constructively across organizational boundaries, must be cultivated through shared training and experience. Their study of Norwegian civil protection exercises reveals that despite shared goals, organizations often enter coordination efforts with differing interpretations, communication norms, and expectations, hindering joint action. Fragmentation arises from limited pre-exercise dialogue, unclear roles, and weak mutual understanding, underscoring the need for joint sensemaking and trust-building activities as part of disaster preparedness. These findings highlight that effective collaboration depends not only on information exchange but also on shared cognition under pressure. In this context, Social Network Analysis (SNA) offers a powerful approach to examine how organizations interact during crises. By mapping relational ties and applying metrics such as centrality, density, and modularity [3, 12], SNA uncovers coordination patterns, identifies key brokers, and reveals potential structural vulnerabilities. Longitudinal SNA further captures the evolution of these networks across disaster phases, providing insight into how collaboration adapts in real-time [1, 9].

Integrating Social Network Analysis with Complex Adaptive Systems (CAS) theory enriches the study of disaster response by highlighting the emergent, adaptive, and decentralized nature of interagency coordination. CAS views disaster response as a self-organizing system where actors adjust to changing conditions

Table 1: Top 5 agencies by Degree and Betweenness Centrality

Node	Degree Centrality	Betweenness Centrality	Closeness Centrality	Eigenvector Centrality
1212	0.8365	0.1292	0.8595	0.2272
538	0.8077	0.1112	0.8387	0.2261
508	0.7885	0.1020	0.8254	0.2249
1663	0.7115	0.0836	0.7761	0.2031
1275	0.6442	0.0618	0.7376	-

**Figure 1: Number of articles published by top organizations during the Cyclone Idai response phase.****Figure 2: Temporal distribution of situation report publications between March and April 2019.**

through feedback and collective learning [6]. This framework is especially relevant for complex crises like Cyclone Idai, where diverse organizations must collaborate under uncertainty. As Elvegård et al. [8] noted, effective coordination relies on adaptive sensemaking rather than rigid hierarchies. CAS enables the analysis of evolving network structures, emphasizing resilience and the reconfiguration of roles over time. Recent studies by Davis et al. [7] and Al & Faisal [2] showed that CAS-informed approaches enhance real-time decision-making and governance agility. In this study, CAS underpins the longitudinal examination of coordination dynamics, offering insights into role shifts, the emergence of key actors, and the formation of modular, functionally aligned structures. This study employs SNA to examine coordination dynamics among humanitarian agencies during Cyclone Idai, a major disaster in southern Africa. Drawing on ReliefWeb situation reports, organization mentions are extracted to build bipartite agency-report networks, which are then projected into agency–agency collaboration graphs. The analysis is temporally segmented into early, mid, and late response phases to trace the evolution of coordination. Key metrics, including degree and eigenvector centrality, Louvain community detection, and type-based assortativity, are used to identify influential actors and structural patterns. The remainder of the paper is organized as follows: Section 2 reviews relevant literature; Section 3 outlines the data and methods; Section 4 presents the results and interpretation; and Section 5 concludes with implications for disaster governance and future research.

2 Literature Review

The academic understanding of inter-organizational collaboration in disaster contexts has evolved from viewing coordination as a procedural necessity to recognizing it as a dynamic social process shaped by institutional diversity and environmental volatility. Elvegård et al. [8] emphasized that effective collaboration in crisis scenarios is not solely dependent on structure but on deliberate design of interaction spaces, enabling shared understanding and

improvisational problem-solving under uncertain conditions. Their work underscores the importance of simulation-based training that supports learning-by-doing, a recognition that traditional models fail to sufficiently replicate real-world complexity [8]. Parallel to this, recent scholarship in archaeology led by Torrence et al. [17] expanded the conceptual foundation of collaboration by illustrating how resilience in past societies depended on distributed networks of agents engaged in adaptive learning and mutual adjustment. These long-term perspectives stress the importance of redundancy, modularity, and decentralized governance in managing unpredictable events, suggesting deep historical roots for what are now considered best practices in contemporary disaster response [17].

Moreover, integrating SNA with CAS theory offers a robust framework to capture the emergent, non-linear nature of coordination. Davis et al. [7] argued that such an integration provides tools to assess organizational behavior not just by position in a network but by responsiveness and adaptability to unfolding situations. Similarly, Al & Faisal [2] demonstrated that CAS-informed SNA helps to identify how local feedback mechanisms and actor-level decisions influence overall network resilience. Together, these studies support a shift from static to dynamic, multi-level analysis of disaster governance. They validate the relevance of studying collaboration not as a fixed configuration of roles but as an evolving ecosystem. This paper builds upon these insights by employing SNA to examine how humanitarian organizations interact during Cyclone Idai, focusing on the temporal evolution of coordination patterns and how structural shifts reflect changing operational priorities.

3 Data Description and Methodology

3.1 Data Collection and Rationale

We collected structured data from ReliefWeb’s public API (RWAPI¹), a curated platform aggregating reports from over 4000 disaster relief organizations. RWAPI provides cleaner, more accurate access

¹ReliefWeb API documentation: [here](https://reliefweb.org/rwapi/)

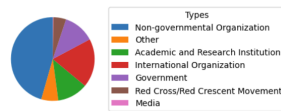


Figure 3: Distribution of Participating Organizations by ReliefWeb Classification.

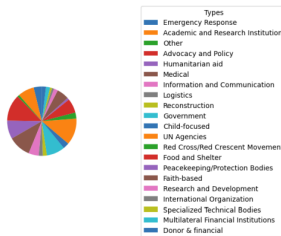


Figure 4: Refined Classification of Participating Organizations by Functional Mandate.

compared to traditional scraping tools like BeautifulSoup or Selenium, allowing filtered queries by disaster type, date, and source². We focused on situation reports related to Cyclone Idai, published between March 1 and April 30, 2019, covering the early, mid, and late response phases. To identify agency participation, metadata on organizations, including names, acronyms, and types, was retrieved from the `/sources` endpoint³. While spaCy-based Named Entity Recognition was initially tested, it underperformed in this context. Instead, we applied a dictionary-based matching of organization names and acronyms across the full report content. This approach improved precision and interpretability, ensuring the accuracy required for reliable network analysis.

3.2 Parsing and Network Construction

To enable reliable analysis, all textual data from reports and organizational metadata were standardized through lowercasing and whitespace normalization. Agency mentions were identified using exact string matching of names and acronyms across report fields, yielding a bipartite dataset that links organizations to the reports in which they appear. To examine temporal shifts in collaboration, the dataset was segmented into three periods, namely early response (March 9–16), mid-term response (March 17–30), and recovery (March 31–April 30). These intervals reflect typical disaster response phases from rescue to stabilization. Bipartite networks were constructed using NetworkX, with nodes representing agencies and reports and edges denoting agency participation. From this, we derived weighted unipartite projections where edges between agencies represent co-occurrence in one or more reports. The resulting networks support both structural and temporal analysis of interorganizational coordination.

²Reports endpoint: <https://api.reliefweb.int/v1/reports>

³Sources endpoint: <https://api.reliefweb.int/v1/sources>

3.3 Analytical Framework

The analysis centers on the projected agency–agency network, where key structural properties were assessed using standard SNA metrics. Degree centrality identified agencies with high collaboration volume, while betweenness centrality highlighted actors bridging different network segments. Eigenvector centrality was used to capture influence through connectedness to other prominent nodes. To detect modular structures, the Louvain algorithm was applied, revealing tightly connected communities that may represent thematic or regional coordination clusters. Organizational types were refined beyond ReliefWeb’s default categories to include UN agencies, medical responders, logistics providers, and advocacy groups, allowing more granular interpretation of sectoral roles. Attribute assortativity was then computed to assess homophily, indicating whether collaboration patterns were sector-specific or cross-sectoral. Network robustness was evaluated through failure simulations, where top central nodes were removed to observe changes in diameter, path length, and connectivity, thus testing structural resilience under stress. Visualization was implemented using NetworkX and Matplotlib, with node size reflecting centrality, color denoting organizational type, and edge thickness representing collaboration frequency. Temporal phases were visualized separately to capture shifts in coordination dynamics. This methodological approach balances interpretability and scalability, enabling a nuanced understanding of interagency collaboration and its evolution during disaster response.

4 Results and Discussion

The coordination network analysis of the Cyclone Idai response uncovered important structural and temporal patterns in interagency collaboration. Utilizing co-occurrence data from 192 curated situation reports, we identified 105 distinct humanitarian organizations, enabling a detailed assessment of coordination dynamics. As shown in Figure 1, OCHA, WFP, and the Logistics Cluster emerged as the most prolific contributors, underscoring their central roles in managing emergency logistics and coordination. Figure 2 illustrates the temporal distribution of reports, with notable peaks in late March and mid-April corresponding to operational milestones such as cholera vaccination campaigns. The original ReliefWeb classification shown in Figure 3 reveals a predominance of non-governmental and international actors, while government and media entities were underrepresented. For greater analytical precision, organizations were reclassified into 18 functional categories as illustrated in Figure 4, offering a clearer view of the sectoral engagement across the response. A bipartite network was then constructed to link agencies with the reports in which they appeared, as seen in Figure 5, and subsequently projected into a unipartite agency–agency network (see Figure 6) to map direct and indirect collaborative ties based on co-reporting patterns.

Centrality analysis of the projected network revealed a concentration of influence among a small set of key actors. Degree and betweenness centrality measures identified UN OCHA, WHO, and UNICEF as the most structurally central agencies, frequently appearing across reports and serving as bridges between otherwise disconnected subgroups. These bridging roles are particularly critical in disaster settings, where timely information exchange across

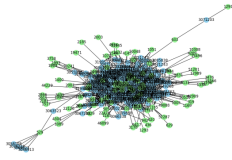


Figure 5: Bipartite network of agencies and reports.

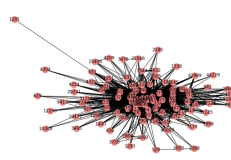


Figure 6: Projected agency-agency collaboration network.

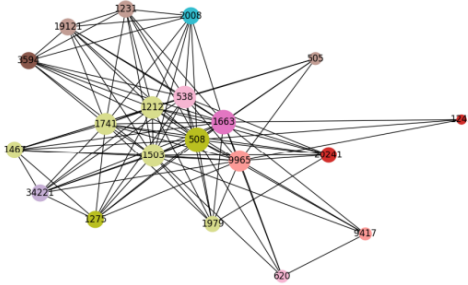


Figure 7: Agency Collaboration Network: Early Phase (March 9–16).

diverse organizational clusters can significantly impact coordination efficiency. The centrality rankings are presented in Table 1, highlighting the dual importance of these agencies in both network density and connectivity. Further validation using closeness and eigenvector centrality, as shown in Table 1, confirmed their prominence, indicating strong access to the wider network and influential positioning among peer organizations. The temporal segmentation of the network into early, mid, and late phases enabled the observation of evolving coordination patterns. During the early phase, spanning March 9 to 16, the network displayed a centralized structure dominated by a few core international actors. This centralized configuration is illustrated in Figure 7, emphasizing rapid leadership deployment and top-down communication. As the response progressed into the mid-term phase (March 17 to 30), decentralization became apparent. Figure 8 shows the emergence of regional and local NGOs, reflecting a shift toward distributed coordination and specialized operational roles.



Figure 8: Agency Collaboration Network: Mid-Term Phase (March 17–30).



Figure 9: Agency Collaboration Network: Recovery Phase (March 31–April 30).

The coordination network exhibited a greater decentralization during the recovery phase (March 31–April 30). As shown in Figure 9, influence became more distributed, with clusters forming around thematically aligned organizations. This reflects a maturation of the

response system, where operational responsibilities diversified and localized leadership gained prominence. The emergence of multiple sub-centers suggests an adaptive coordination architecture capable of supporting parallel operations while maintaining indirect ties with global actors. Edge weights, representing collaboration frequency, declined over time. In the early phase, thick and dense edges among central agencies indicated intensive coordination, which is typical in the immediate aftermath of disasters. As the response progressed, ties became more fragmented, with a broader set of weaker links. While this dispersion supports inclusivity and flexibility, it also increases the risk of coordination breakdowns without centralized communication protocols or shared operational tools. A granular reclassification of organizations revealed functional clusters centered on mandates such as health, logistics, and protection, indicating that thematic alignment shaped collaboration more than organizational type. Louvain-based community detection confirmed this modular structure, reinforcing the importance of domain-specific coordination. However, the limited presence of national government agencies highlights a data bias in international reporting, underscoring a common limitation in SNA-based disaster studies, informal, or undocumented collaborations remain invisible. Though efficient, the persistent core-periphery structure observed in early stages may marginalize peripheral actors and restrict horizontal knowledge flow. Addressing such imbalances through joint training programs, standardized communication frameworks, and inclusive coordination platforms could strengthen multilevel collaboration. Overall, this study demonstrates that disaster coordination networks evolve from centralized to distributed forms. SNA, combined with thematic analysis, offers critical insights for designing resilient and adaptive response systems.

5 Conclusion

This study demonstrates the effectiveness of SNA in examining inter-agency coordination during disaster response, using Cyclone Idai as a case study. The evolving network structure highlighted shifts from centralized to distributed coordination, reflecting the adaptive involvement of both global and local actors. Recent works, such as Lv et al. [15], have shown how dynamic SNA can track trust and responsiveness in health emergencies, while Chen et al. [4] emphasize decentralized coordination in flood response. Building on these findings, future work should explore multimodal data fusion, real-time monitoring, and role-based network optimization to enhance coordination effectiveness in complex emergencies.

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